# TYLER W. GRAY, MS

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*PhD Candidate, Pharmaceutical Sciences | Pharmacology & ADME/Tox | PK/PD & DMPK | Scientific AI Model Evaluation*

## PROFILE

Pharmaceutical sciences PhD candidate specializing in pharmacology, ADME/Tox, and PK/PD, with three years of early-stage drug discovery experience across oncology, CNS, and psychiatric programs. Originating inventor on three novel drug candidates. Interpret bioactivity, binding kinetics, and compound liability data to drive lead selection, and curate and verify drug discovery datasets at scale across ChEMBL, BindingDB, and UniProt. Serve as a chemistry and biology domain expert evaluating frontier AI model outputs, authoring benchmark tasks, and writing grading rubrics for scientific data quality. Expertise spans mechanism of action, GPCR pharmacology, structure activity relationships, liposomal drug delivery, IND-enabling study design, and NIH grant writing.

## PHARMACY & CLINICAL EXPERIENCE

**Automation Systems Coordinator (Pyxis ES)** | UCHealth Anschutz Medical Campus | 2021 – 2023

- Managed hospital wide automated dispensing infrastructure for inpatient pharmacy operations serving oncology, critical care, and surgical units at a major academic medical center.
- Monitored performance metrics to identify override rate anomalies and dispensing error patterns; implemented data driven workflow interventions achieving USP 797/800 and Joint Commission compliance.
- Trained pharmacists, nurses, and clinical staff on Pyxis ES and Medstation 4000 systems; coordinated software updates and technology integrations across pharmacy and clinical departments.

**Retail Pharmacy Technician** | Kroger Pharmacy | 2008 – 2012

- Interpreted and processed medication orders and prescriptions in a high-volume retail pharmacy; maintained records, inventory, and medication organization for dispensing accuracy.

## AI MODEL EVALUATION & SCIENTIFIC BENCHMARKING

**AI Safety Expert Writer, Chemistry & Biology Domain** | Mercor (Contract) | 2025 – Present

- Serve as a chemistry and biology domain expert red-teaming frontier AI models, testing whether model outputs correctly separate legitimate scientific inquiry from requests carrying genuine misuse potential.
- Design grading rubrics and benchmark task sets across three escalating risk tiers, authoring technically credible prompts that probe model reasoning at each level of complexity and hazard.
- Evaluate model responses against a defined policy standard, diagnose failures in scientific logic, factual accuracy, and calibration, and document both over-refusal and under-refusal error modes in written feedback.
- Write reference answers with full technical reasoning, establishing the ground-truth benchmarks used to score model outputs and improve downstream training data quality.

## RESEARCH EXPERIENCE

**PhD Researcher (Small Molecule Biophysics & Drug Discovery)** | University of Colorado Anschutz | 2023 – Present

- Generate and interpret protein-ligand binding kinetics on a LUMICKS C-Trap HF optical tweezer system, extracting KD values, dwell times, and force-dependent unbinding profiles at single-molecule resolution that resolve population heterogeneity invisible to ensemble SPR and TR-FRET readouts.
- Apply dwell time survival analysis and exponential decay modeling to extract kinetic rate constants from kymograph time series, producing binding thermodynamic parameters informing lead candidate selection across three active drug discovery programs.
- Author NIH R21 grant application for a receptor-targeted liposomal Alzheimer's platform, developing specific aims, preliminary data framing, and IND-adjacent regulatory narrative connecting particle engineering parameters to clinical endpoints.
- Conduct SAR analysis, mechanism of action assessment, and molecular docking with AutoDock Vina and Avogadro for scaffold optimization, pairing each series with ADME liability screening, CYP interaction assessment, and hERG cardiac risk triage to prioritize lead candidates across active programs, including Foseralink™.

## ENTREPRENEURIAL EXPERIENCE

**Founder & CEO** | Krush Productions Cultivation | 2014 – 2019

- Founded and led a cultivation enterprise with 15+ employees, P&L responsibility, B2B sales development, and quarterly projection analysis across a multi-client portfolio.
- Developed standard operating procedures, compliance protocols, and client relationship management systems driving sustained B2B revenue growth.

**Real Estate Broker Associate** | Shorewood Real Estate | 2011 – 2021

- Managed full transaction lifecycle for residential and commercial property transfers under Colorado unified broker licensure.

**Founder** | Nuclyx™ - Nuclear Remediation & Energy Lifecycle Technology | 2024 – Present

- Founder of a patent-pending deep-tech venture developing novel materials science approaches to nuclear waste remediation and energy lifecycle advancement; proprietary technology details available under NDA.
- Designed a multi-component separation and encapsulation system targeting radioisotope containment and environmental remediation, integrating advanced membrane engineering and nanomaterial science.
- Evaluating Delaware C-corp entity structure for institutional investment readiness; developing commercialization roadmap targeting nuclear energy sector decarbonization and waste lifecycle reduction goals.

## DRUG DISCOVERY AND NOVEL CANDIDATE RESEARCH PROJECTS

### Foseralink™ - SSRI-Folate Hybrid Prodrug for Treatment-Resistant Depression

Conceived and designed a hybrid prodrug architecture delivering both pharmacophores through a single cleavable-linker molecule. Led the DMPK assessment, covering linker stability profiling across simulated gastric and intestinal pH, metabolic activation and cleavage kinetics, and ADME liability screening of the intact prodrug prior to activation. These data informed the decision to retain the cleavable-linker design over a fixed-dose combination and established the safety rationale for the intact species. Addresses co-morbid folate deficiency and serotonergic dysregulation through synchronized dual pharmacophore release at SERT and FOLR1.

### Liposomal CNS Delivery Systems (NIH R21)

- Designed a receptor-targeted pegylated liposome formulation which is currently the basis of an NIH R21 grant proposal with specific aims covering biophysical characterization, in vitro BBB transcytosis validation in a transwell endothelial model, and preliminary in vivo CNS biodistribution.
- Originating inventor of a pegylated liposomal formulation for advancing the IND-enabled pharmacology package per FDA drug product guidance.

### Receptor Bioactivity SAR Analysis

Extracted and normalized IC50, Ki, and EC50 bioactivity data for GPCR ligands across ChEMBL and BindingDB, reconciling values reported under differing assay formats, species, and readout conventions. Cross-referenced UniProt accessions to identify receptor subtype conflation and isoform misassignment, then applied primary literature verification to adjudicate each discrepancy and build a curated SAR dataset for scaffold optimization.

## TECHNICAL SKILLS

### Assay Science & Biophysics

- LUMICKS C-Trap HF optical tweezers
- Single-molecule biophysics
- Force spectroscopy & kymograph analysis
- KD determination & binding kinetics
- Protein-ligand thermodynamics
- Dwell time survival analysis
- Prodrug & cleavable linker chemistry
- Assay interpretation & QC
- Dose-response & Hill analysis
- Cross-assay data normalization

### Bioactivity Databases

- ChEMBL / BindingDB / UniProt / PubChem

### Drug Discovery & Formulation

- SAR & lead optimization
- Mechanism of action analysis
- Target & cellular pharmacology
- Liposomal / nanoparticle formulation
- Pegylated liposome design (EPR targeting)
- Compound optimization planning
- AutoDock Vina / Avogadro / molecular docking

### ADME/Tox & DMPK

- ADME liability screening
- hERG & cardiac liability triage
- CYP inhibition & drug-drug interaction

### PK/PD & Regulatory

- PK/PD modeling (linear & non-linear)
- CYP enzyme pharmacology / DDI
- Michaelis-Menten & compartmental PK
- NIH R21 grant writing
- IND-enabling study design
- USP 797 / USP 800
- FDA liposomal drug product guidance
- ICH guidelines / 21 CFR Part 11

### Clinical Systems

- Pyxis ES / Medstation 4000
- Automated dispensing optimization
- Medication safety & error reduction
- USP 797/800 sterile compounding

- Protein target verification
- Bioactivity data curation
- UniProt accession mapping
- Receptor subtype annotation
- Primary literature review & annotation
- Metabolic activation & cleavage kinetics
- pH-dependent stability profiling
- BBB permeability & transcytosis
- Structural alert identification
- Compound liability assessment
- Clinical-IT cross-functional collaboration
- Technical documentation & SOP authoring

### AI & Data Tools

- AI / LLM model evaluation
- Grading rubric & benchmark design
- Model reasoning error diagnosis
- RLHF & scientific data annotation
- Prompt engineering
- Salesforce CRM
- Microsoft Office Suite
- EPIC EHR / ePRN systems

## EDUCATION

**Doctor of Philosophy (PhD)**
*Pharmaceutical Sciences (Candidate)*
University of Colorado Anschutz, Skaggs School of Pharmacy and Pharmaceutical Sciences
Expected 2027

**Master of Science (MS)**
*Biomedical Science*
University of Maryland, School of Pharmacy
2022

**Bachelor of Arts (BA)**
*Chemistry / Biology*
Metropolitan State University of Denver
2012